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Question 1:

```
#include <iostream>
```

```
#include <vector>
```

```
#include <string>
```

```
using namespace std;
```

```
class Student{
```

```
    private:
```

```
        int rollNumber;
```

```
        string name;
```

```
        double percentage;
```

```
        static int totalStudents;
```

```
    public:
```

```
        Student(int r,string n)
```

```
        {
```

```
            rollNumber = r;
```

```
            name = n;
```

```
            percentage=0;
```

```
            totalStudents;
```

```
            cout<<"[Student Created] Roll:"<<r<<";Name:"<<n<<endl;
```

```
        }
```

```
        Student(const Student &s)
```

```
        {
```

```
            rollNumber = s.rollNumber;
```

```

        name = s.name;
        percentage = s.percentage;
        totalStudents++;
        cout << "[Copy Constructor Called]" << endl;
    }

    ~Student(){
        totalStudents--;
        cout << "[Student Deleted] Roll:" << rollNumber << endl;

    }

    void setPercentage(double p)
    {
        if(p >= 0 && p <= 100)
        {
            percentage = p;
        }
    }

    double getPercentage() const{
        return percentage;
    }

    char calculateGrade() const{
        if(percentage >= 80)
            return 'A';
        else if(percentage >= 70)
            return 'B';
        else if(percentage >= 60)
            return 'C';
    }

```

```
    else if(percentage>=50)
        return 'D';
    else
        return 'F';
}
```

```
bool operator==(const Student &s)
{
    return rollNumber == s.rollNumber;
}
```

```
friend ostream& operator<<(ostream &out,const Student &s)
{
    out<<"Roll"<<s.rollNumber<<"| Name : "<<s.name<<" |Percentage : 
"<<s.percentage<<"% |Grade : "<<s.calculateGrade();

    return out;
}
```

```
static int getTotalStudents()
{
    return totalStudents;
}
```

```
};
```

```
int Student::totalStudents =0;
```

```
int main()
{
```

```

vector<Student> v;

v.emplace_back(101,"Aniket Sawant");
v.emplace_back(102,"Aditay Sawant");
v.emplace_back(103,"Harsh Dudhal ");
v.emplace_back(104,"Tejas Sawant");
v.emplace_back(105,"omkar jadhav");


v[0].setPercentage(86.6);
v[1].setPercentage(75);
v[2].setPercentage(85.6);
v[3].setPercentage(42.65);
v[4].setPercentage(65.62);


cout<<"\nStudents:\n";
for(Student&s : v )
{
    cout<<s<<endl;
}


Student copy = v[0];


if(v[0] == v[1])
    cout<<"\nSame Roll Number\n";
else
    cout<<"\nDifference Roll Number";


int high=0,low =0;


for(int i=1;i<v.size();i++)
{
    if(v[i].getPercentage()>v[high].getPercentage())

```

```
high =i;
```

```
if(v[i].getPercentage()<v[low].getPercentage())
```

```
low =i;
```

```
}
```

```
cout<<"\nHighest : "<<v[high]<<endl;
```

```
cout<<"\nLowest : "<<v[low]<<endl;
```

```
cout<<"\nTotal Students : "
```

```
<<Student::getTotalStudents()<<endl;
```

```
int A=0,B=0,C=0,D=0,F=0;
```

```
for(Student&s:v)
```

```
{
```

```
    switch(s.calculateGrade())
```

```
    {
```

```
        case 'A':
```

```
            A++;
```

```
            break ;
```

```
        case 'B':
```

```
            B++;
```

```
            break;
```

```
        case 'C':
```

```
            C++;
```

```
            break;
```

```
        case 'D':
```

```
            D++;
```

```
            break;
```

```
        case 'F':  
            F++;  
            break;  
        }  
    }
```

```
    cout<<"\n Grade Distribution : \n";\  
    cout<<"A : " <<A<<endl;  
    cout<<"B : " <<B<<endl;  
    cout<<"C : " <<C<<endl;  
    cout<<"D : " <<D<<endl;  
    cout<<"F : " <<F<<endl;  
    return 0 ;  
}
```

Question1.cpp:64:14: **note:** try making the operator a 'const' member function

```
[Student Created] Roll:101,Name:Aniket Sawant
[Student Created] Roll:102,Name:Aditay Sawant
[Copy Constructor Called]
[Student Deletedt] Roll:101
[Student Created] Roll:103,Name:Harsh Dudhal
[Copy Constructor Called]
[Copy Constructor Called]
[Student Deletedt] Roll:101
[Student Deletedt] Roll:102
[Student Created] Roll:104,Name:Tejas Sawant
[Student Created] Roll:105,Name:omkar jadhav
[Copy Constructor Called]
[Copy Constructor Called]
[Copy Constructor Called]
[Copy Constructor Called]
[Student Deletedt] Roll:101
[Student Deletedt] Roll:102
[Student Deletedt] Roll:103
[Student Deletedt] Roll:104
```

Students:

```
Roll101| Name : Aniket Sawant |Percentage : 86.6% |Grade : A
Roll102| Name : Aditay Sawant |Percentage : 75% |Grade : B
Roll103| Name : Harsh Dudhal |Percentage : 85.6% |Grade : A
Roll104| Name : Tejas Sawant |Percentage : 42.65% |Grade : F
Roll105| Name : omkar jadhav |Percentage : 65.62% |Grade : C
[Copy Constructor Called]
```

Difference Roll Number

```
Highest : Roll101| Name : Aniket Sawant |Percentage : 86.6% |Grade : A
```

[Student Deletedt] Roll:103

[Student Deletedt] Roll:104

Students:

Roll101| Name : Aniket Sawant |Perecentage : 86.6% |Grade : A

Roll102| Name : Aditay Sawant |Perecentage : 75% |Grade : B

Roll103| Name : Harsh Dudhal |Perecentage : 85.6% |Grade : A

Roll104| Name : Tejas Sawant |Perecentage : 42.65% |Grade : F

Roll105| Name : omkar jadhav |Perecentage : 65.62% |Grade : C

[Copy Constructor Called]

Difference Roll Number

Highest : Roll101| Name : Aniket Sawant |Perecentage : 86.6% |Grade : A

Lowest : Roll104| Name : Tejas Sawant |Perecentage : 42.65% |Grade : F

Total Students : 1

Grade Distribution :

A :2

B :1

C :1

D :0

F :1

[Student Deletedt] Roll:101

[Student Deletedt] Roll:101

[Student Deletedt] Roll:102

[Student Deletedt] Roll:103

[Student Deletedt] Roll:104

[Student Deletedt] Roll:105

Question 2:

```
#include <iostream>

#include <string>

using namespace std;

class Employee{

private:

    int empld;

    string name;

    double salary;

    int performanceR;

public:

    Employee(int id,const string& empName,double sal,int
rating):empld(id),name(empName),salary(sal),performanceR(rating){

        cout<<"Para construcctor\n";

    }

    double calculateBonus()const{

        if(performanceR==5)

            return salary*0.20;

        else if(performanceR==4){

            return salary*0.15;

        }else if(performanceR==3)

            return salary*0.10;

        else if(performanceR==2){

            return salary*0.05;

        }else

            return 0 ;

    }

}
```

```

void displayDetails() const{

    cout<<"ID : "<<empld

        <<"|Name : "<<name

        <<"|Salary : "<<salary

        <<"|Rating: "<<performanceR

        <<"|Bonus : "<<calculateBonus()

        <<endl;

}


double getBonus()const{

    return calculateBonus();

}

string getName()const{

    return name;

}

};

int main(){

    Employee emp[5]={

        Employee(1,"Aniket Sawant",50000,5),

        Employee(2,"Aditya chormule",35000,4),

        Employee(3,"Om   Sawant",25000,3),

        Employee(4,"Sahil Sawant",15000,2),

        Employee(5,"Harsh Sawant",15000,2),

    };

    for(int i=0;i<5;i++)

    {

        emp[i].displayDetails();

```

```

}

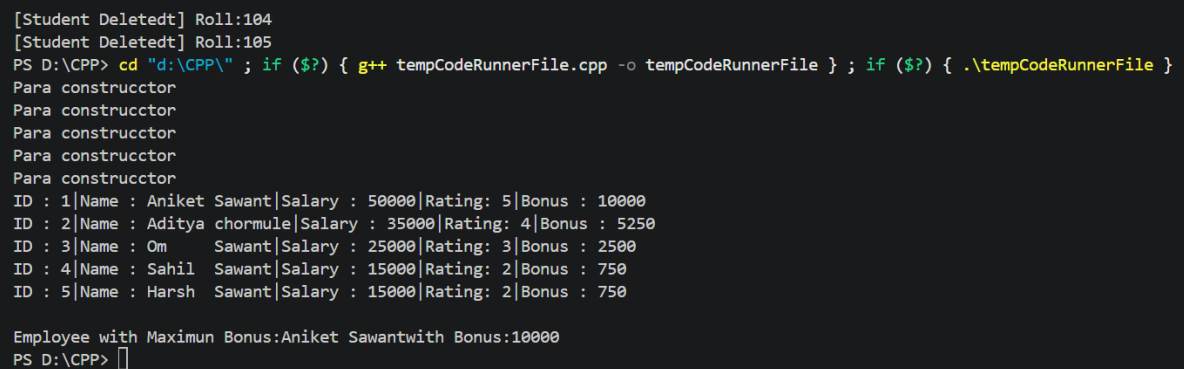
int maxIndex =0;

for(int i=1;i<5;i++)
{
    if(emp[i].getBonus()>emp[maxIndex].getBonus())
    {
        maxIndex=i;
    }
}

cout<<"\nEmployee with Maximun Bonus:"
    <<emp[maxIndex].getName()
    <<"with Bonus:"
    <<emp[maxIndex].getBonus()
    <<endl;

return 0 ;
}

```



```

[Student Deletedt] Roll:104
[Student Deletedt] Roll:105
PS D:\CPP> cd "d:\CPP\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Para constructor
Para constructor
Para constructor
Para constructor
Para constructor
ID : 1|Name : Aniket Sawant|Salary : 50000|Rating: 5|Bonus : 10000
ID : 2|Name : Aditya chormule|Salary : 35000|Rating: 4|Bonus : 5250
ID : 3|Name : Om Sawant|Salary : 25000|Rating: 3|Bonus : 2500
ID : 4|Name : Sahil Sawant|Salary : 15000|Rating: 2|Bonus : 750
ID : 5|Name : Harsh Sawant|Salary : 15000|Rating: 2|Bonus : 750

Employee with Maximun Bonus:Aniket Sawantwith Bonus:10000
PS D:\CPP> 

```

Question 3 :

```
#include <iostream>

using namespace std;

class Temperature{
    private:
        double celsius;

    public:
        Temperature(double c):celsius(c){

        }

        double toFahrenheit() const{
            return (celsius*9/5)+32;
        }
        double toKelvin() const{
            return celsius+273.15;
        }

        void displayAll() const{
            cout<<"Temperature : "<<celsius<<"C = "
                <<toFahrenheit()<<"F = "
                <<toKelvin()<<"K"<<endl;
        }
};

int main()
{
    double celsius;
```

```

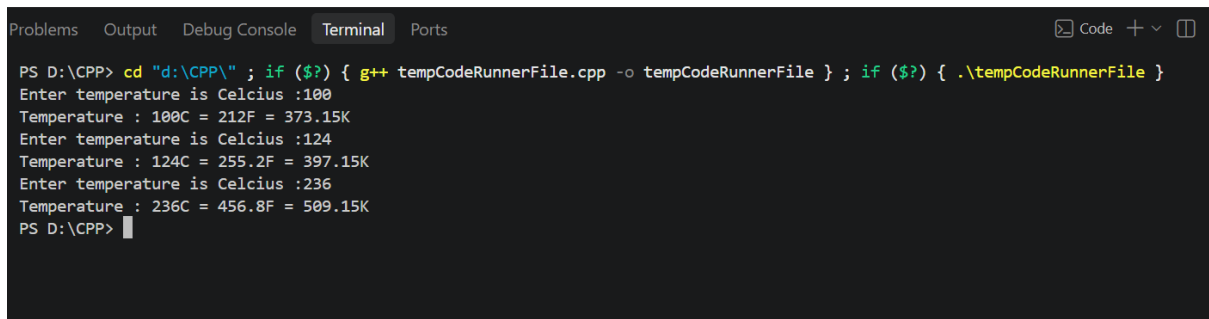
for(int i=0;i<3;i++)
{
    cout<<"Enter temperature is Celcius :";
    cin>>celsius;

    if(celsius<-273.15)
    {
        cout<<"ERROR : Temperature below absolute zero(-273.15C)"<<endl;
    }else{
        Temperature t(celsius);
        t.displayAll();
    }

}

return 0;
}

```



```

PS D:\CPP> cd "d:\CPP\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter temperature is Celcius :100
Temperature : 100C = 212F = 373.15K
Enter temperature is Celcius :124
Temperature : 124C = 255.2F = 397.15K
Enter temperature is Celcius :236
Temperature : 236C = 456.8F = 509.15K
PS D:\CPP>

```